

W007 — CI/CD Pipeline Plan

Wave: W007

Status: Planned. A ready-to-use workflow file already exists in the repo as `ci/minerva-pid-test.yml` because the connected GitHub App currently lacks the `workflows` permission and therefore cannot push files under `.github/workflows/`. W007's first job is to land that workflow properly.

Goal: make the MINERVA P&ID pipeline continuously verifiable — every push/PR regenerates derived artifacts and runs the full standalone test battery, with deterministic, reviewable results.

1. Constraints carried from earlier waves

These are non-negotiable and shape every CI decision:

- **No pytest.** Tests are standalone runners (`_run_all()` + `if __name__ == "__main__"`), with `_data_available()` skip-guards for data-dependent suites. CI must invoke each `tests/test_*.py` directly.
- **Stdlib-first.** Runtime deps are minimal; `cairosvg` (PDF render) needs system `libcairo2`. CI must install that OS package.
- **Reproducibility via `./make.sh`.** CI regenerates everything from sources in `data/svg/` and then runs the tests — it must not rely on committed derived artifacts.
- **Known nondeterminism.** `reports/COMPONENT_CATALOG_v2.xlsx` differs by ~1 byte between runs (zip timestamp/ordering), which cascades into `reports/W005_validation_report.md` size. CI must tolerate or normalize this (see §4).
- **No auto-merge.** CI reports status; humans merge.

2. Pipeline stages

```
checkout → setup matrix (Python) → apt: libcairo2 → pip: requirements
→ ./make.sh (regenerate all derived outputs)
→ run every tests/test_*.py (aggregate pass/fail)
→ golden-file diff gate (statistics JSONs)
→ upload artifacts (publish/*.html, reports/*.md, data/crossmap/*.json)
```

2.1 Build/regenerate

- Run `./make.sh` end-to-end. It must exit 0. This is itself a strong integration test (7 steps: model → W003/W004 → catalog → W005 → **W006 crossmap** → atlas v6 → collage → **W006 viewer**).

2.2 Test

- Loop over `tests/test_*.py`, run each with `PYTHONPATH=src python3 <file>`, capture exit codes, fail the job if any returns non-zero.
- Current battery (44 assertions): `test_colour_model` (5), `test_integration_pipeline` (5), `test_w003_w004` (10), `test_w005_reconciliation` (11), `test_w006_crossmap` (13).

2.3 Golden-file gate (cross-pollinated pattern)

- Treat the committed `reports/*_statistics.json` as golden snapshots. After `make.sh`, diff regenerated stats against committed ones; **fail on semantic drift** (counts changing) but **ignore** the known XLSX byte jitter.
- This is the same golden-file discipline used in the test suites, lifted to CI level.

3. Matrix & environment

Axis	Values	Rationale
Python	3.10, 3.11, 3.12	repo targets stdlib that is stable across these; catches version drift
OS	ubuntu-latest	libcairo2 available via apt

System setup step:

```
sudo apt-get update && sudo apt-get install -y libcairo2
python -m pip install --upgrade pip
pip install -r requirements.txt # cairosvg, openpyxl, pyyaml (if pinned)
```

4. Handling the known nondeterminism

Two viable strategies (pick one in implementation):

1. **Normalize before diff (preferred):** strip the file-size column from `W005_validation_report.md` and exclude `*.xlsx` from the golden gate, comparing only semantic JSON stats.
2. **Tolerance gate:** allow `±2 bytes` on the single XLSX/size line, fail on anything larger.

CI must **never** commit regenerated artifacts back (no write-back), avoiding noisy diffs.

5. The `workflows` permission blocker (must resolve first)

- The Abacus GitHub App connection used for pushes **cannot write** `.github/workflows/*.yaml` (missing `workflows` scope). PR #547 worked around this by committing the workflow as `ci/minerva-pid-test.yaml` plus documentation.
- **W007 action items:**
 1. Ask the maintainer to grant the `workflows` permission to the [Abacus GitHub App](https://github.com/apps/abacusai/installations/select_target) (https://github.com/apps/abacusai/installations/select_target), **or** have a maintainer manually `git mv ci/minerva-pid-test.yaml .github/workflows/minerva-pid-test.yaml` and push.
 2. Once landed, verify the Actions run is green on a throwaway PR.
 3. Add a status badge to the MINERVA P&ID README.

6. Deliverables / Definition of done

1. `.github/workflows/minerva-pid-test.yml` active on `main` (resolves \$5).
 2. Matrix build green on Python 3.10–3.12.
 3. `./make.sh` runs in CI and exits 0.
 4. All `tests/test_*.py` pass in CI (44+ assertions).
 5. Golden-file gate active and tolerant of the documented XLSX jitter.
 6. Build artifacts (viewer HTML, reports, crossmap JSON) uploaded per run.
 7. Status badge in README.
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7. Future (post-W007)

- Cache pip + apt to speed runs.
- Optional: publish `publish/*.html` to GitHub Pages for always-fresh interactive viewer + atlas.
- Optional: scheduled (nightly) run to catch environment drift.